

INTRODUCTION TO MACHINE LEARNING

Basic Concepts, Supervised Learning, Unsupervised Learning

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2026-I



Outline

- 1 Fundamentals of Machine Learning
- 2 Supervised Machine Learning
- 3 Unsupervised Machine Learning



Outline

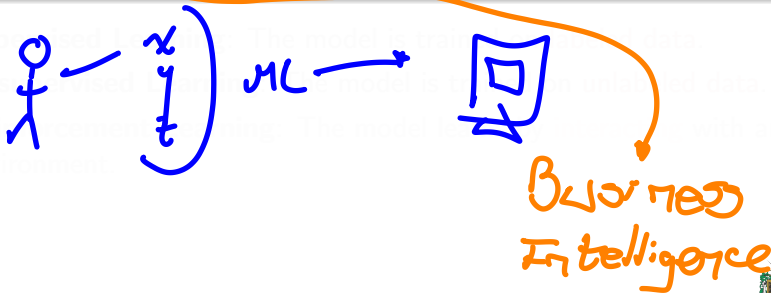
- 1 Fundamentals of Machine Learning
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Key Concepts in Machine Learning

Machine Learning

- **Machine learning** is a method of data analysis that automates analytical model building.
- It is a **branch** of **artificial intelligence** based on the idea that systems can learn from data, identify patterns and make decisions with minimal human intervention.



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- **Supervised Learning:** The **model** is trained on **labeled data**.

- **Unsupervised Learning:** The model is trained on **unlabeled data**.
- **Reinforcement Learning:** The model learns by **interacting** with an environment.

X (features)

1 2
3 5
4 9
2 7
...

8
13
9

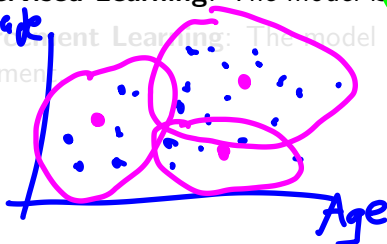


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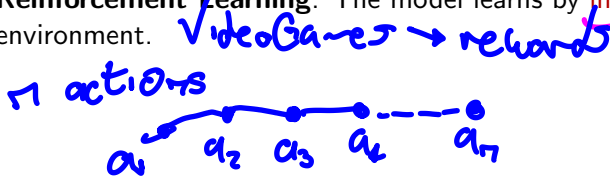
Any Grade



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Typical Machine Learning Problems

- **Classification:** Predicting a label.

Supervised

- **Regression:** Predicting a continuous value.

- **Clustering:** Grouping similar data points.

- **Dimensionality Reduction:** Reducing the number of features.

- **Anomaly Detection:** Identifying unusual data points.

- **Association Rule Learning:** Identifying relationships between variables.

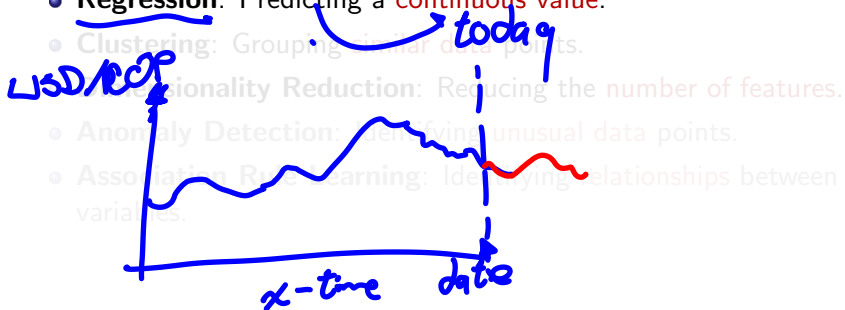
semi-supervised

unsup. $\Rightarrow$ Group



Typical Machine Learning Problems

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salary *amount* *not*
x₀ *x₁* *x₂*
y₀ *y₁* *y₂*



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200 - 300 - 400

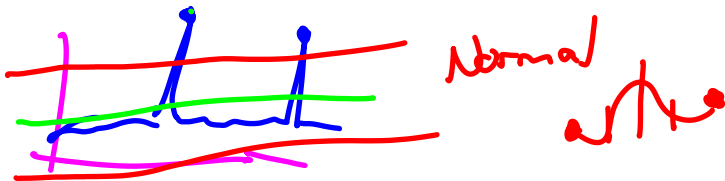
1 feature - 1 column



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Outliers



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The Machine Learning Workflow

- **Data Collection:** Gathering the data

Excel

PDFs

DB (Rel
NoSQL)

API

⋮
Multimedia

① Historical
(? - today)

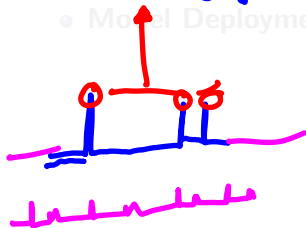
② Fresh data
↓
Periodically



The Machine Learning Workflow

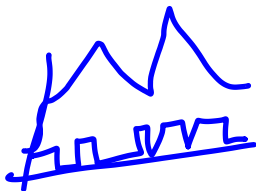
- **Data Collection:** Gathering the **data**.
- **Data Preprocessing:** Cleaning and preparing the **data**.
- **Feature Engineering:** Creating new features.

Null?
outliers?



format

- lowercase names
- dates Y-M-d
- drift R:00:00



The Machine Learning Workflow

- **Data Collection:** Gathering the **data**.
- **Data Preprocessing:** Cleaning and preparing the **data**.
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Semester	Start	Delay
8	2020-2	↓
7	2022-1	↓
9	2021-2	○
1	2025-1	↓
2	2025-2	○



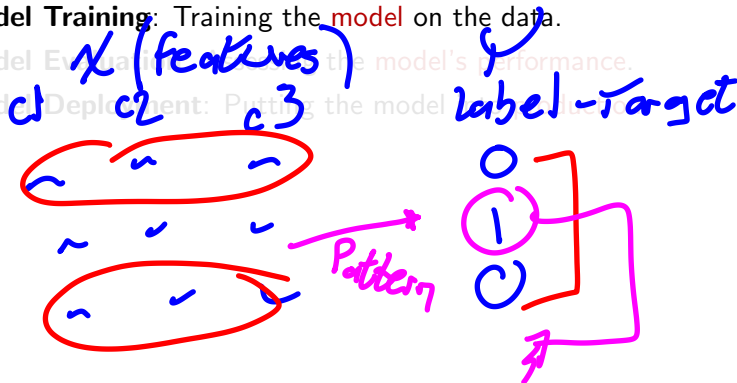
The Machine Learning Workflow

- **Data Collection:** Gathering the **data**.
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- **Model Selection:** Choosing the **best model**.
- **Model Training:** Training the **model** on the data.
- **Model Evaluation:** Assessing the **model's performance**.
- **Model Deployment:** Putting the model into production.



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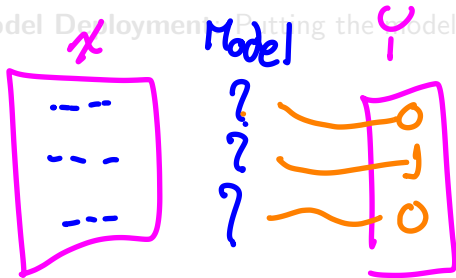
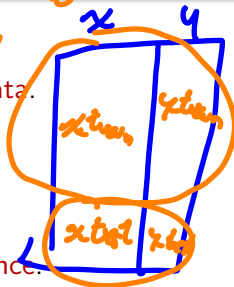


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0.7-0.8

Train



Good ↑
Bad ↓

Validate
(test)



The Machine Learning Workflow

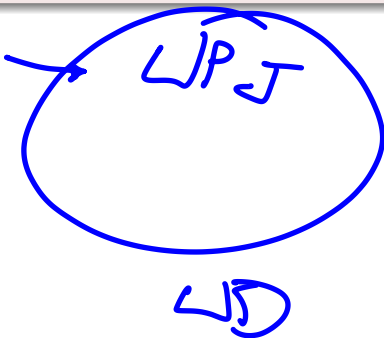
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Algorithmic Bias

- **Algorithmic bias** is a systematic error in a model that results in unfair outcomes.
- It can be caused by biased training data, biased algorithms, or biased decision-making.

Distr. Pizza



RAG



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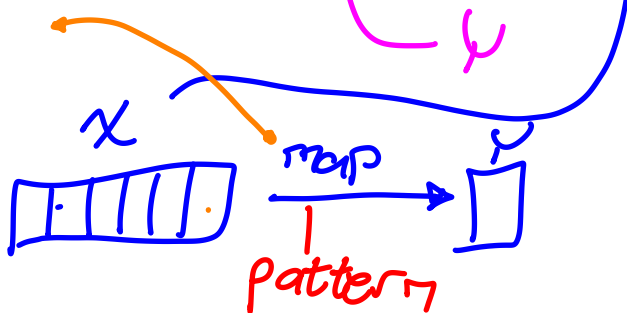


Introduction to Supervised Machine Learning

Definition

- **Supervised learning** is a type of **machine learning** where the model is trained on **labeled data**.
- It involves **training** a model to **map input data to output data** based on example **input-output pairs**.

$f(x,y)$
 $\downarrow \text{map}$
 $f(\theta, r)$



Overfitting and Underfitting

Overfitting

Overfitting occurs when a model learns the training data too well and performs poorly on new data.

Underfitting

Underfitting occurs when a model is too simple to capture the underlying structure of the data.

$$2 + 2 = 4$$

Train

$$7 + 4 = 11$$

$$29992 + 99532 = ?$$

Test

	Train	80	20	Test
3	—	2	1	
4	—	3	1	
9	—	4	2	
10	—	8	2	



Overfitting and Underfitting



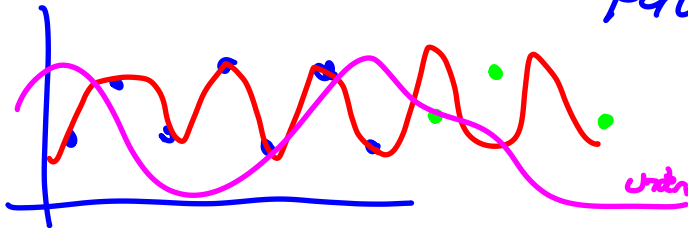
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No Pattern

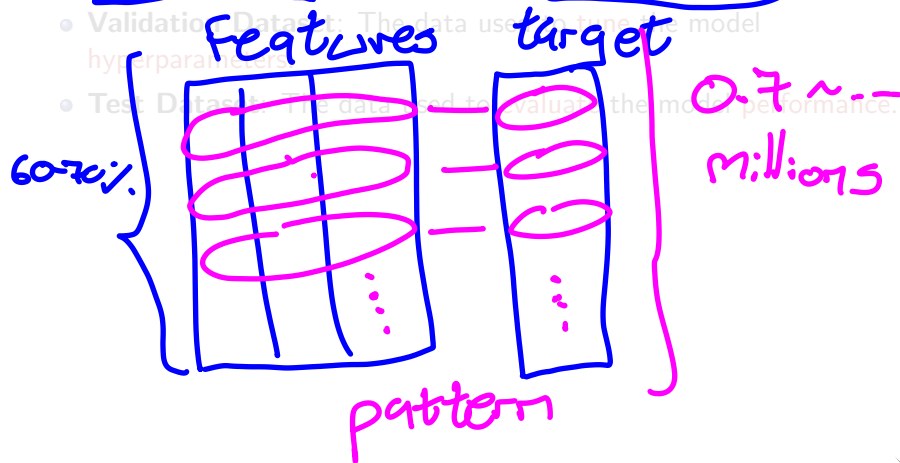


Supervised Learning Datasets

- **Training Dataset:** The data used to **train the model**.

- **Validation Dataset:** The data used to tune the model hyperparameters.

- **Test Dataset:** The data used to evaluate the model performance.

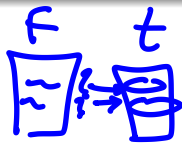


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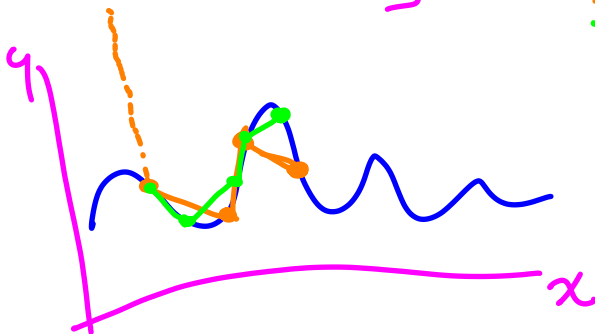
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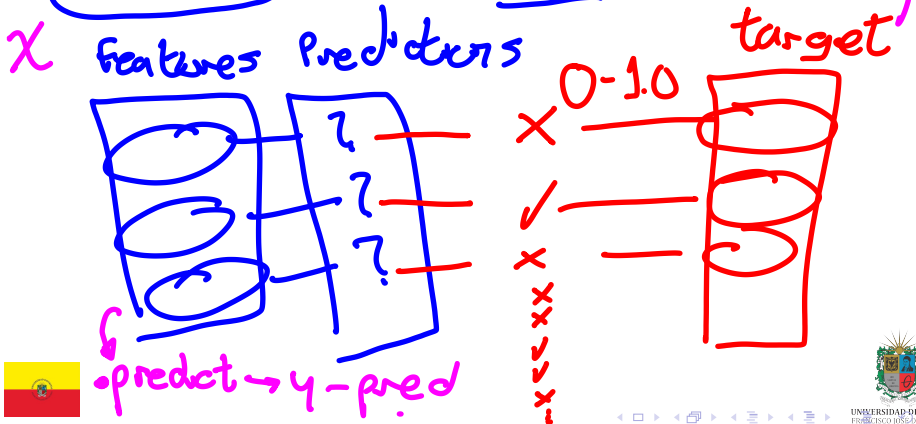
the ratio is

learning rate $\Rightarrow$ dist

$f(x) = \dots$



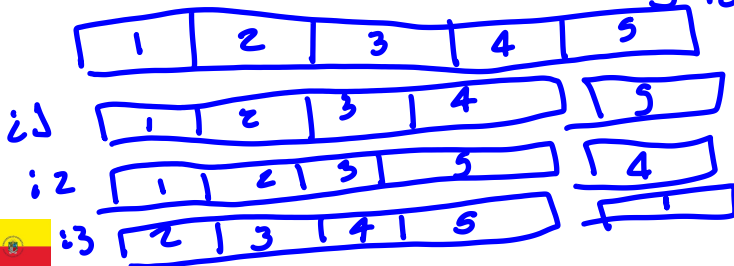
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Cross-Validation

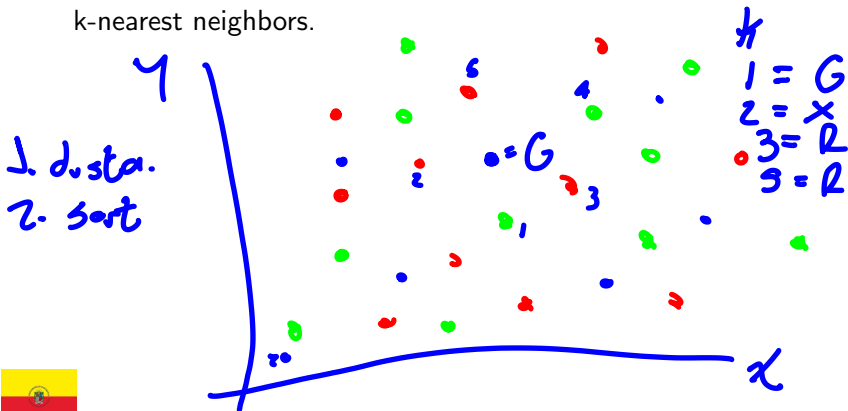
- **Cross-validation** is a technique for assessing the performance of a model.
- It involves splitting the data into multiple subsets, training the model on some subsets, and evaluating it on others.
- Common cross-validation **techniques** include k-fold cross-validation and leave-one-out cross-validation.
- Cross-validation helps to reduce overfitting and provides a more accurate estimate of the model's performance.

$$5 \text{ folds} \Rightarrow k=5$$



K-Nearest Neighbors

- **K-Nearest Neighbors (KNN)** is a simple algorithm that stores all available cases and classifies new cases based on a **similarity measure**.
- It can be used for both **classification** and **regression** tasks.
- For **classification**, the output is the **class label** of the majority of the k-nearest neighbors.



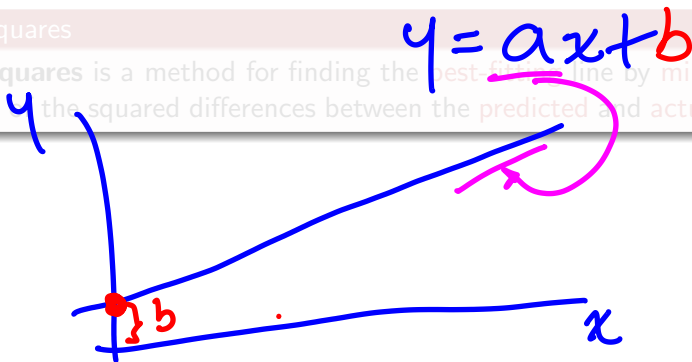
Linear Regression with Least Squares

Linear Regression

- Linear regression is a type of regression analysis used for predicting the value of a continuous dependent variable.
- It works by finding the line that best fits the data.

Least Squares

Least squares is a method for finding the best-fitting line by minimizing the sum of the squared differences between the predicted and actual values.



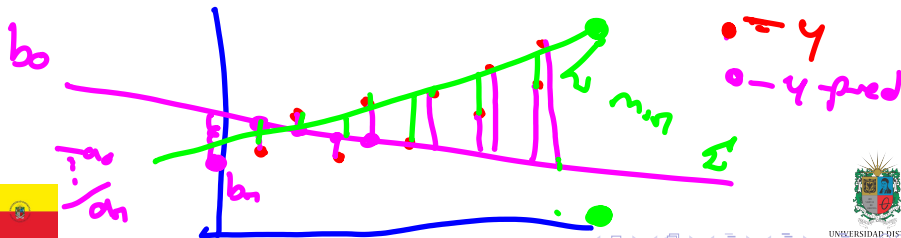
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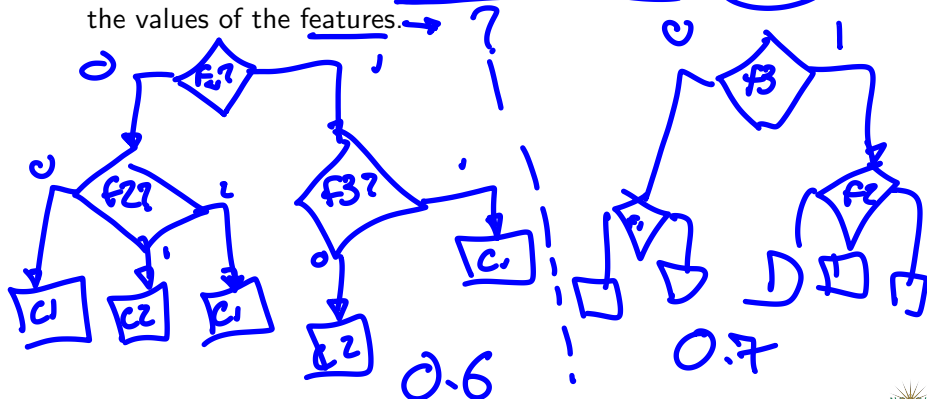
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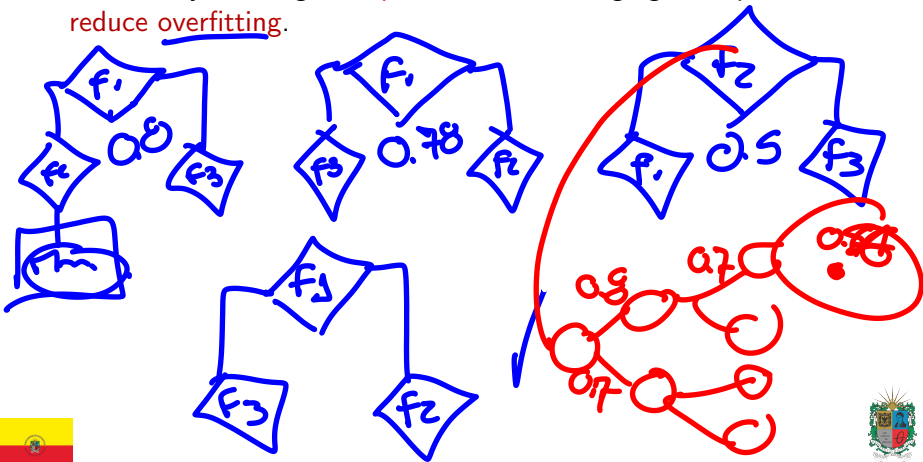
Decision Trees

- **Decision trees** are a type of **machine learning model** that can be used for both **classification** and **regression** tasks.
- They work by recursively **partitioning** the data into **subsets** based on the values of the **features**.



Random Forest

- **Random forest** is an **ensemble learning** method that combines **multiple decision trees** to create a strong predictive model
- It works by building **multiple trees** and averaging their predictions to **reduce overfitting**.



Neural Networks

- **Neural networks** are a type of **machine learning model** inspired by the **human brain**.
- They consist of **layers** of interconnected nodes that process **input data** and produce **output data**.



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Introduction to Unsupervised Machine Learning

Definition

- **Unsupervised learning** is a type of **machine learning** where the model is trained on **unlabeled data**.
- It involves finding **patterns** and **relationships** in the data without any predefined labels.



Clustering

- **Clustering** is a type of **unsupervised learning** that involves **grouping** similar **data points** together.
- Common clustering algorithms include **k-means**, **hierarchical clustering**, and **DBSCAN**.



K-means

- **K-means** is a popular **clustering algorithm** that partitions the data into K **distinct clusters** based on feature similarity.
- It works by iteratively assigning data points to the nearest **cluster centroid** and updating the centroids based on the assigned points.



Anomaly Detection

- **Anomaly detection** is a type of **unsupervised learning** that involves identifying **unusual data points** in a dataset.
- Common anomaly detection algorithms include **Isolation Forest**, **One-Class SVM**, and **Autoencoders**.



Autoencoders

- **Autoencoders** are a type of **neural network** used for unsupervised learning.
- They work by **encoding** the input data into a lower-dimensional representation and then **decoding** it back to the original data.



Dimensionality Reduction

- **Dimensionality reduction** is a technique for **reducing** the number of **features** in a dataset while retaining as much information as possible.
- Common dimensionality reduction techniques include **Principal Component Analysis (PCA)** and **t-Distributed Stochastic Neighbor Embedding (t-SNE)**.



Principal Component Analysis (PCA)

- **Principal Component Analysis (PCA)** is a **statistical technique** used for **dimensionality reduction**.
- It works by transforming the data into a **new coordinate system** where the greatest variance lies along the first principal **component**, the second greatest variance along the second principal **component**, and so on.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/machine-learning>

